

NATIONAL TRANSPORTATION SAFETY BOARD

Investigative Hearing

US Army PAT25, Sikorsky UH-60L and

PSA Airlines flight 5342

MHI RJ Aviation CL-600-2C10 (CRJ700), N709PS

Midair Collision in Washington, DC

January 29, 2025

Docket No.	SA-544
EXHIBIT	
12-HELO-A	

Cockpit Voice Recorder (Helicopter)

Factual Report

(116 Pages)

National Transportation Safety Board

Office of Research and Engineering

Washington, DC 20594



DCA25MA108

COCKPIT VOICE RECORDER (HELICOPTER)

Group Chairman's Factual Report

July 14, 2025

WARNING

The reader of this report is cautioned that the summary and transcript of a cockpit voice recorder audio recording is not a precise science but is the best product possible from a National Transportation Safety Board group investigative effort. The summary and transcript or parts thereof, if taken out of context, could be misleading. The summary and transcript should be viewed as an accident investigation tool to be used in conjunction with other evidence gathered during the investigation. Conclusions or interpretations should not be made using the summary and transcript as the sole source of information.

TABLE OF CONTENTS

A. ACCIDENT.....	3
B. COCKPIT VOICE RECORDER GROUP	3
C. DETAILS OF THE INVESTIGATION	3
1.0 RECORDER DESCRIPTION.....	4
1.1 Recorder Condition	4
1.2 Audio Recording Description.....	7
1.3 Timing and Correlation	7
1.4 Description of Audio Events.....	8
APPENDIX A. CVR QUALITY RATING SCALE.....	9
APPENDIX B. CVR TRANSCRIPT	10

A. ACCIDENT

Location: Washington, D.C.
Date: January 29, 2025
Time: 20:48 eastern standard time (EST)
Helicopter: Sikorsky UH-60L, US Army, Tail No. 00-26860, Priority Air Transport (callsign PAT25)
Airplane: Mitsubishi Heavy Industries (MHI Regional Jet (RJ) Aviation (formerly Bombardier) CL-600-2C10 (CRJ700), PSA Airlines Flight 5342, N709PS (callsign Bluestreak 5342)

B. COCKPIT VOICE RECORDER (HELICOPTER) GROUP

Group Chairman	Charles Cates Mechanical Engineer/Recorder Specialist National Transportation Safety Board (NTSB)
Group Member	Mark Ward Aviation Accident Investigator NTSB
Group Member	CW3 Bryan Caudle Maintenance Officer/Pilot US Army
Group Member	Edwin Miller Senior Accident Investigator Federal Aviation Administration (FAA)

C. DETAILS OF THE INVESTIGATION

A CVR group was convened on February 3, 2025 to develop the summary and transcript, and reconvened on May 15, 2025 to transcribe additional air traffic control radio calls audible to the crew. The FAA did not participate in the reconvened group on May 15.

The NTSB Vehicle Recorder Division received the following cockpit voice and data recorder (CVDR):

Recorder Manufacturer/Model:	Goodrich Integrated Vehicle Health Management Unit (IVHMU) (Penny & Giles MPFR)
Part Number:	051631
Recorder Serial Number:	A00528-005

1.0 Recorder Description

The US Army Sikorsky UH-60L helicopter is equipped with a Goodrich Integrated Vehicle Health Management Unit (IVHMU) recorder. Integrated as a part of the IVHMU is a Penny & Giles Multi-Purpose Flight Recorder (MPFR) Cockpit Voice and Flight Data Recorder (CVFDR).

The Penny & Giles Multi-Purpose Flight Recorder (MPFR) is a solid-state recorder capable of simultaneously recording up to four channels of digital audio. Each channel retains the most recent 120 minutes of operation. In the UH-60L installation only two channels are recorded, one from each pilot's audio panel. The other two channels are unused.

1.1 Recorder Condition

The recorder was recovered from the helicopter while it was under water and immediately transported to the NTSB Vehicle Recorder laboratory. Initial inspection of the IVHMU upon arrival at the Vehicle Recorder laboratory showed no evidence of impact or fire damage (see figures 1 and 2). The IVHMU was disassembled and the MPFR was accessed. The MPFR was in good condition but was wet.

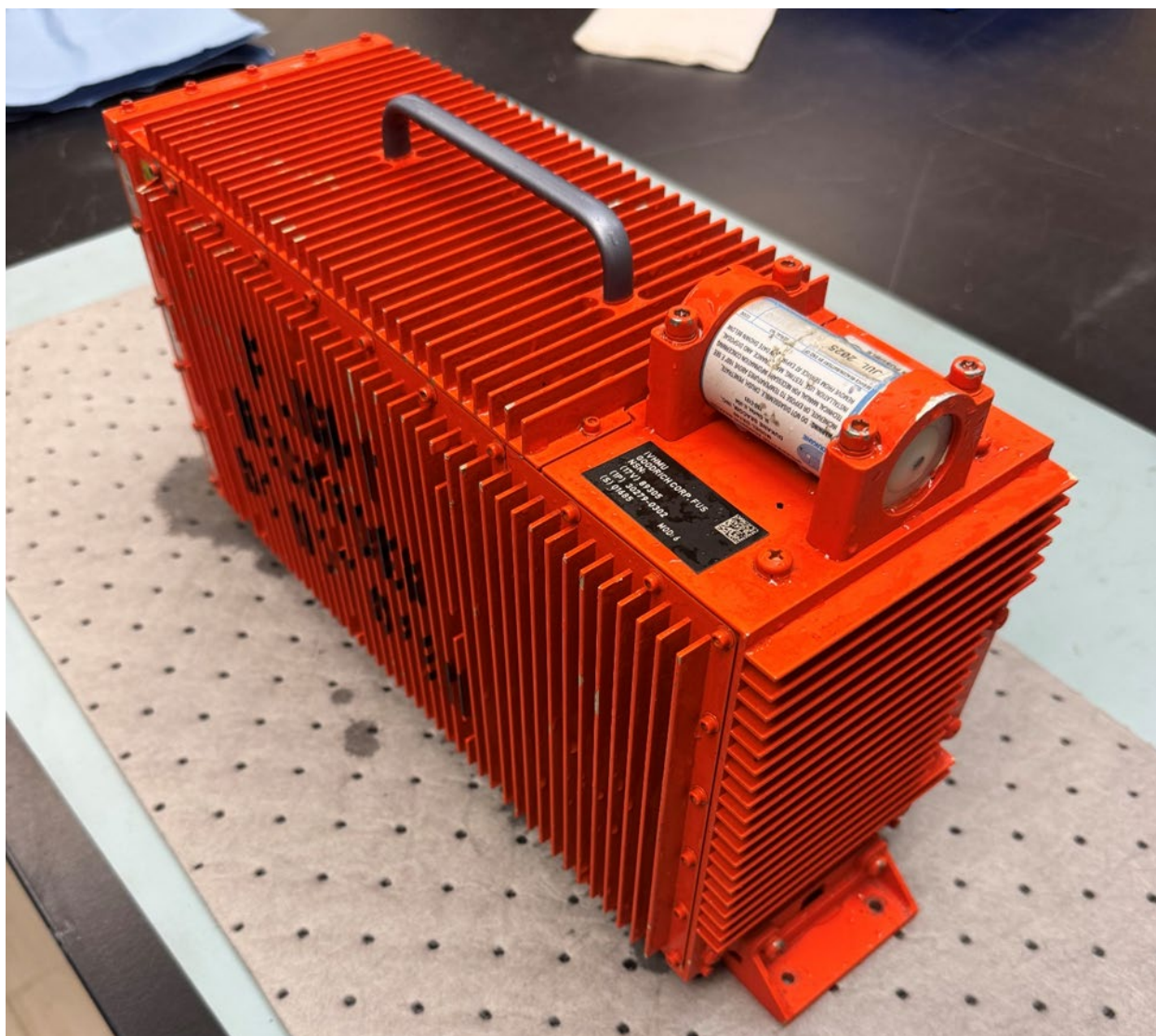


Figure 1. IVHMU, as received by the Vehicle Recorder Laboratory



Figure 2. IVHMU data plate.

The MPFR uses a crash survivable design made up of several cylinders to protect the memory contained within. The solid-state memory boards are called the crash survivable memory module panel electric circuit (CSMM PEC). The CSMM PEC is contained within a cylinder known as the "core", which is housed inside of the main cylinder called the Crash Box Assembly (see figure 3).



Figure 3. Crash box assembly data plate.

The Crash Box Assembly was removed from the IVHMU and opened to reveal the core (see figure 4). Water had ingressed into the Crash Box Assembly along the flexible circuit memory ribbon cable and the core was found to be wet.



Figure 4. MPFR core with exposed flexible circuit memory ribbon cable and connector after removal from the Crash Box Assembly.

The core was cleaned and dried and electrical checks were performed via the exposed CSMM PEC connector. Based on those checks, the CSMM PEC was found to be intact. The CSMM PEC was successfully downloaded using NTSB's equipment and the recorder manufacturer's software.

1.2 Audio Recording Description

The MPFR can record up to four audio channels, but the installation on the UH-60L only utilizes two audio channels. One is from the pilot internal communication system (ICS) panel and the second is from the copilot ICS panel. The helicopter's ICS recorded both the push-to-talk intercom for communications between crew members including pilot, co-pilot, and crew chief, and the monitored air traffic control radio frequency traffic. Intra-crew communications are only recorded when the ICS push-to-talk button is depressed. Radio traffic is always recorded, even when the intercom push-to-talk is depressed. The MPFR also recorded aural alerts of some critical aircraft systems. There is no cockpit area microphone (CAM) installed on the aircraft. Each channel's audio quality is indicated in Table 1.¹

Table 1. Audio Quality

Channel Number	Content/Source	Quality	Duration
1	Pilot ICS	Excellent (Intercom)/ Poor (Radio)	121 minutes
2	Copilot ICS	Excellent (Intercom)/ Poor (Radio)	121 minutes

1.3 Timing and Correlation

Timing on the transcript was established by correlating common air traffic control transmissions recorded by both the CRJ and the helicopter. Because the CRJ recorder had timing information, this could be used to set the time of the helicopter recording. Specifically, radio transmissions from Washington tower that were

¹ Appendix A comprises the CVR Quality Rating Scale.

recorded by both aircraft were used to establish the offset from recorder elapsed time to local eastern standard time (EST) of the recording.

1.4 Description of Audio Events

The recording began at 18:46:52, with the helicopter in flight. The helicopter's callsign was PAT two five. During the first twenty-four minutes of the recording, the instructor pilot (IP) was the pilot flying and the pilot being evaluated was navigating by pilotage utilizing a time distance heading card. The IP did not note any deficiencies or make any corrections of the pilot being evaluated during this period.

Throughout the recording the crew commented on the winds and the resulting effects of turbulence and ground speed of the helicopter. Radio transmissions received from ground stations and other helicopters were generally broken and of poor quality. The transcript began at 19:11:08, as the helicopter was on approach to a landing zone at Greenhouse Airport (02VA) in Stevensburg, VA.

For most of the transcribed portion of the recording, air traffic control radio transmissions were only transcribed if they were addressed to PAT25. Beginning when the crew checked in with Potomac approach at 20:29:41, all air traffic control radio transmissions audible in the ICS were transcribed. Transmissions from air traffic control that occurred while the crew was talking to each other on the helicopter intercom were denoted in italics.

The recording ended at 20:48:04.

Submitted by:

Charles Cates
Mechanical Engineer/Recorder Specialist

APPENDIX A. CVR QUALITY RATING SCALE

The levels of recording quality are characterized by the following traits of the cockpit voice recorder information:

Excellent Quality Virtually all of the crew conversations could be accurately and easily understood. The transcript that was developed may indicate only one or two words that were not intelligible. Any loss in the transcript is usually attributed to simultaneous cockpit/radio transmissions that obscure each other.

Good Quality Most of the crew conversations could be accurately and easily understood. The transcript that was developed may indicate several words or phrases that were not intelligible. Any loss in the transcript can be attributed to minor technical deficiencies or momentary dropouts in the recording system or to a large number of simultaneous cockpit/radio transmissions that obscure each other.

Fair Quality The majority of the crew conversations were intelligible. The transcript that was developed may indicate passages where conversations were unintelligible or fragmented. This type of recording is usually caused by cockpit noise that obscures portions of the voice signals or by a minor electrical or mechanical failure of the CVR system that distorts or obscures the audio information.

Poor Quality Extraordinary means had to be used to make some of the crew conversations intelligible. The transcript that was developed may indicate fragmented phrases and conversations and may indicate extensive passages where conversations were missing or unintelligible. This type of recording is usually caused by a combination of a high cockpit noise level with a low voice signal (poor signal-to-noise ratio) or by a mechanical or electrical failure of the CVR system that severely distorts or obscures the audio information.

Unusable Crew conversations may be discerned, but neither ordinary nor extraordinary means made it possible to develop a meaningful transcript of the conversations. This type of recording is usually caused by an almost total mechanical or electrical failure of the CVR system.

APPENDIX B. CVR TRANSCRIPT

The following is a transcript of a Penny & Giles MPFR CVR, serial number A00528-005, installed on a US Army Sikorsky UH-60L, Tail No. 00-26860, Priority Air Transport callsign PAT25 involved in an accident on January 29, 2025 in Washington, D.C.

LEGEND

APR-P	Radio transmission from Potomac approach
BLJK1	Radio transmission from helicopter callsign Blackjack 1
FPS	Aural alert from the helicopter's Flight Path Stabilization System
INT	Flight crew intercom audio panel voice or sound source
ARCR1	Radio transmission from helicopter callsign AirCare 1
MUSL12	Radio transmission from helicopter callsign Muscle 12
MUSL13	Radio transmission from helicopter callsign Muscle 13
MUSL7	Radio transmission from helicopter callsign Muscle 7
RDO	Radio transmission from PAT25
TWR-A	Radio transmission from the DCA airport tower controller
-1	Voice identified as the instructor pilot
-2	Voice identified as the pilot being evaluated
-3	Voice identified as the helicopter crew chief
-?	Voice unidentified
*	Unintelligible word
#	Expletive
@	Non-pertinent word
()	Questionable insertion
[]	Editorial insertion

Note 1: Times are expressed in EST.

Note 2: Generally, only air traffic control radio transmissions to and from the accident aircraft were transcribed. Beginning when the crew checked in with Potomac approach at 20:29:41, all air traffic control radio transmissions audible in the helicopter ICS were transcribed.

Note 3: Words shown with excess vowels, letters, or drawn-out syllables are a phonetic representation of the words as spoken.

Note 4: A non-pertinent word, where noted, refers to a word not directly related to the operation, control, or condition of the aircraft.

Note 5: Radio transmissions from Washington Tower *presented in italics* were audible to the crew in the ICS but occurred simultaneously to other intra-aircraft communications between the crew.

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
18:46:52.0			
	START OF RECORDING		
19:11:08			
	START OF TRANSCRIPT		
19:11:08.9			
INT-2	alright get the before landing we're gonna hopefully be wheels down in the next like seven or eight minutes.		
19:11:14.9			
INT-2	and we are locked parking brake released crew passenger mission equipment secure.		
19:11:18.3			
INT-3	left rear secure.		
19:11:19.3			
INT-2	* * .		
19:11:20.6			
INT-2	alright there's another reservoir passing over now this is also on the map this is great actually we're in a good spot if we're crossing this one.		
19:11:39.1			
INT-1	@[Crew Chief] do you need heat back there bud?		
19:11:41.9			
INT-3	what's that?		
19:11:42.7			
INT-1	do you need heat back there?		
19:11:44.7			
INT-3	uh heat?		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:11:46.0 INT-1	yeah.		
19:11:46.5 INT-2	turn the heat on.		
19:11:47.8 INT-3	sure. I'll take a little bit. you guys can set it to like low.		
19:12:07.9 INT-3	thank you.		
19:12:10.7 INT-1	you're—.		
19:12:28.1 INT-2	ok we should see some wires kinda coming up on the north side.		
19:12:38.6 INT-2	let me know if we cross that road to the west we've gone too far.		
19:12:42.4 INT-1	rog-o.		
19:13:22.4 INT-2	we've lost a little bit of ground speed you can pick up the pace a little bit.		
19:13:25.3 INT-1	say again?		
19:13:26.2 INT-2	we've lost a bit of ground speed you can bring up the pace a hair.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:13:29.1 INT-1	alright. speedin' up.		
19:13:30.7 INT-2	back to ninety five or so.		
19:13:31.0 INT-1	eight hundred feet and a hundred knots indicated.		
19:13:34.7 INT-2	thank you.		
19:13:43.1 INT-2	* water tower looks familiar.		
19:14:03.1 INT-2	the river's snaking up to the right and the road's snaking up to the left.		
19:14:06.6 INT-1	rog-o.		
19:14:08.6 INT-2	that's good. come right a little bit. another...		
19:14:12.2 INT-1	right.		
19:14:14.2 INT-2	...three two five or so.		
19:14:17.3 INT-1	right to three two five.		
19:14:38.6 INT-2	pick up the speed a little bit back to a hundred ground.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:14:42.1 INT-1	speedin' up.		
19:14:59.9 INT-1	comin' left on course.		
19:15:01.7 INT-3	clear left.		
19:15:27.1 INT-2	should be about there.		
19:15:30.5 INT-2	back to the right.		
19:15:32.9 INT-1	comin' right.		
19:15:36.5 INT-2	and we're gonna change like a three two zero. or three three zero.		
19:15:40.9 INT-3	clear left.		
19:15:44.3 INT-1	alright. three three zero.		
19:15:45.9 INT-2	yup.		
19:15:48.3 INT-2	and pick up the speed a little bit more.		
19:15:51.7 INT-1	* I got it.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:15:55.1 INT-1	keep turnin' me in to it.		
19:15:57.6 INT-2	turning you in to the light or something? *		
19:15:59.8 INT-1	no to the wind.		
19:16:01.3 INT-2	yeah.		
19:16:02.1 INT-1	I'll speed up and then I turn a little bit and I lose it all.		
19:16:12.6 INT-1	are we R-P inbound at this point?		
19:16:14.6 INT-2	we should be.		
19:16:23.1 INT-2	final heading is gonna be three one zero so you can turn to that.		
19:16:28.7 INT-1	okay comin' left.		
19:16:45.1 INT-2	we know it's pretty close to the road.		
19:16:48.1 INT-1	roger searchin' for the road. got the road on the left movin' left to right.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:17:03.5 INT-2	little more speed.		
19:17:06.1 INT-1	speedin' up.		
19:17:08.4 INT-2	five k. some of these little ponds are familiar.		
19:17:48.8 INT-2	crossing another road.		
19:17:51.5 INT-1	roger.		
19:17:55.7 INT-2	that large lit building may be part of it.		
19:17:59.1 INT-1	it do look like a greenhouse.		
19:18:00.5 INT-2	yeah it does doesn't it.		
19:18:02.6 INT-2	we can start slowin' back.		
19:18:04.4 INT-1	alright slowin' back.		
19:18:06.9 INT-2	about...		
19:18:10.4 INT-2	(there you) go.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:18:12.1 INT-2	the final ultimate landing heading should be about a three three zero.		
19:18:19.5 INT-1	I can definitely say I have the greenhouse in sight.		
19:18:21.2 INT-2	yeah ok.		
19:18:22.1 INT-1	it is blinding the # out of me.		
19:18:23.6 INT-2	[sound of quick laugh] go ahead I think we have this long pathway here to the right if we need to go around and set back down we can.		
19:18:32.1 INT-1	yeah I don't- I'm not seein' anything that looks like a landing area. where you at ma'am? where are you looking at?		
19:18:37.0 INT-2	just uh- yeah just right of the greenhouse let's see.		
19:18:43.4 INT-2	down a bit- I think you just passed it. let's see.		
19:18:49.0 INT-2	so bright.		
19:18:51.1 INT-1	yeah that's uh pretty damn bright.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:18:52.9 INT-2	I thought that might have been the building that was in the pictures but I don't think it was.		
19:18:58.2 INT-2	expecting it to be lit at night. that's intense.		
19:19:01.1 INT-1	that is ridiculous.		
19:19:05.6 INT-2	keep slowin' back I think we're right about at the landing area.		
19:19:10.1 INT-1	sorry I'm just trying to get over that thing. yeah I think I've got the strip out the left door. and clear left.		
19:19:16.9 INT-2	okay. clear left.		
19:19:18.3 INT-1	comin' left.		
19:19:34.0 INT-1	alright. gonna roll out here for a moment. wind's still at our right front. number one needle's pointing back to it.		
19:19:44.3 INT-2	'kay.		
19:19:47.9 INT-1	maintaining six hundred feet @[Crew Chief] for your situational awareness two hundred fifty feet over the ground.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:19:53.1 INT-3	roger.		
19:19:54.1 INT-1	and clear on left.		
19:19:55.1 INT-3	clear left.		
19:20:12.8 INT-1	got that tower there. no factor for us.		
19:20:34.6 INT-1	alright startin' my approach so where should I be ma'am?		
19:20:39.7 INT-2	off to the right. the greenhouse (out/at) your left door.		
		19:20:43.3 RDO-?	[sound similar to seven radio clicks, consistent with operating field pilot controlled lighting (PCL)]
19:20:45.6 INT-1	the greenhouse should be out my left door?		
19:20:48.9 INT-2	I think it's off to...that's the strip too so come right a little further.		
19:20:54.0 INT-1	comin' right.		
19:20:54.3 INT-2	oh sorry left. I'm so sorry. I'm-I totally * there's the tower that's the runway that's the strip.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:20:59.6 INT-1	where's the strip? I need you to point it out to me.		
19:21:00.9 INT-2	it's right here. okay.		
19:21:01.8 INT-1	you take flight controls.		
19:21:03.0 INT-1	you've got the flight controls.		
19:21:03.0 INT-2	I've got the flight controls.		
19:21:04.5 INT-2	alright there we go.		
19:21:08.4 INT-1	alright I have the landing strip in sight.		
19:21:10.2 INT-2	yeah me too.		
19:21:12.0 INT-2	got that little tower no factor.		
19:21:19.8 INT-1	alright we are in to the wind very well so I like that.		
19:21:22.4 INT-2	yeah okay great.		
19:21:24.4 INT-2	* * feet.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:21:24.5 INT-1	we are locked released crew passengers mission equipment.		
19:21:28.0 INT-3	left rear secure.		
19:21:28.9 INT-2	* building.		
19:21:33.3 INT-2	descent.		
19:21:38.6 INT-3	clear of obstacles clear on the left.		
19:21:41.1 INT-2	thank you coming forward.		
19:21:42.6 INT-1	clear down right.		
19:21:43.9 INT-2	nice and wide.		
19:21:49.0 INT-1	fifty feet.		
19:21:50.9 INT-2	*.		
19:21:54.5 INT-2	thirty feet.		
19:21:57.6 INT-1	looks nice and dry.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:21:59.2 INT-2	yup. looks pretty flat. minor grass. not worried about it.		
19:22:17.5 INT-2	no slope to speak of. pressure's out. and handle. * * *		
19:22:22.0 INT-1	alright cool.		
19:22:24.4 INT-1	handle's up. parking brake set.		
19:22:25.3 INT-2	cool.		
19:22:25.7 INT-1	uh @[Crew Chief] if you want to try and fix our door.		
19:22:28.0 INT-3	yup I'll hop out and fix that.		
19:22:28.8 INT-2	yeah thank you.		
19:22:30.8 INT-1	alright um...timing's a little off its pretty windy.		
19:22:34.6 INT-2	yeah.		
19:22:35.2 INT-1	uh and I think we were here at about the right time.		
19:22:39.1 INT-2	pretty close.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:22:41.6 INT-1	uh yeah so.		
19:22:43.6 INT-1	alright I'll take back flight controls.		
19:22:45.2 INT-2	alright you can have flight controls.		
19:22:46.5 INT-1	I have the controls.		
19:22:49.2 INT-1	yeah we were pretty close on it. and then just had to uh come around. that greenhouse is really...		
19:22:54.7 INT-2	it's really bright.		
19:22:55.8 INT-3	alright. it's locked.		
19:22:56.5 INT-2	thanks @[Crew Chief].		
19:22:57.2 INT-3	yup.		
19:22:57.9 INT-2	have you been down here at night and seen that?		
19:22:59.0 INT-1	no.		
19:23:00.0 INT-2	that's crazy.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:23:00.9 INT-1	yeah that thing's uh pretty...pretty detrimental.		
19:23:04.8 INT-2	yeah I would've uh had us avoid that a little bit. sorry about that.		
19:23:10.8 INT-1	all good.		
19:23:18.5 INT-2	okay I can close out my fuel check.		
19:23:21.2 INT-1	roger.		
19:23:22.0 INT-2	*		
19:24:18.4 INT-2	* eleven fifty.		
19:24:22.2 INT-1	alright eleven fifty burn rate.		
19:24:23.3 INT-2	yup and I'm going to go ahead and do the auto mode for our transfer. verify that it's working. yup.		
19:24:33.8 INT-2	*.		
19:24:38.1 INT-1	I do see fuel transfer.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:24:40.5 INT-2	I agree with you. and we'll keep that monitored for any imbalance *.		
19:24:54.2 INT-3	left rear is back inside and re-secured.		
19:24:56.6 INT-1	rog-o.		
19:24:58.0 INT-2	sweet thanks.		
19:25:02.4 INT-2	okay. and coming out of here...we'll be...avoiding...two J um...		
19:25:16.4 INT-2	that's Culpeper I think K-J-R		
19:25:18.3 INT-1	K-C-J-R yup		
19:25:19.2 INT-2	or K-C-J-R yup.		
19:25:20.7 INT-1	cool pepper.		
19:25:22.9 INT-2	coolest pepper.		
19:25:28.2 INT-1	yeah I'm going to use that strip more. make people fly into that # greenhouse.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:25:32.2 INT-2	yeah yeah.		
19:25:39.0 INT-1	I actually don't know if that strip is still in the F-A-A.		
19:25:43.4 INT-2	that's a fair point.		
19:25:44.4 INT-1	I was trying to look it up last night and it wouldn't show up. looks like they didn't renew some paperwork or something.		
19:25:49.3 INT-2	okay.		
19:25:51.0 INT-2	I found them on uh flight aware or something like googling but yeah didn't check the uh...didn't check it. okay alright so taking out of here with this heading which * to be a little off it looks like it's more like a three three five no three four three.		
19:26:09.7 INT-1	cool.		
19:26:10.5 INT-2	* we can use this to finish taking off the runway get over the trees and then we'll turn to- expect a- two two zero. back myself up with my T-D-H.		
19:26:30.2 INT-2	so yeah to the- yeah probably to the two zero two zero.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:26:37.0 INT-1	so right turn over to a zero two zero?		
19:26:39.5 INT-2	coo-rect. yup.		
19:26:41.8 INT-2	play with the lighting a little bit.		
19:26:43.5 INT-1	yeah.		
19:26:46.0 INT-2	that should get us up to Warrenton. and in the mean time I'm gonna switch over to * Quantico * stay with them but I'll switch over to Potomac here pretty soon.		
19:26:57.8 INT-1	rog-o.		
19:27:39.3 INT-1	@[Crew Chief] do you have a preferred method for this transfer?		
19:27:43.2 INT-3	what do you mean?		
19:27:44.2 INT-1	uh well a lot of guys especially down in Hondo and prolly just cause we were so far from anything um they would want to transfer up to like twenty three and then turn it off and do another fuel check. I'm fine just leaving it on auto. it's whatever your comfort level is.		
19:27:57.9 INT-3	yeah I'm fine leaving it on auto.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:27:59.4 INT-1	okay perfect. yeah I'm good with that. it's just yeah I know some guys like to fill it turn it off redo a fuel check continue to do that which is... not a bad thing either.		
19:28:02.4 INT-2	kinda makes sense.		
19:28:18.0 INT-3	nah I usually just * with my first burn rate work with that until I exhaust the externals and then compute with and then start a fuel check with remaining internal fuel.		
19:28:28.3 INT-1	yeah that kind of was always my take on it is it's gonna keep us keep us up there and then when it's burned out do another fuel check for the final twenty two hundred or whatever.		
19:28:35.5 INT-3	yup.		
19:28:42.3 INT-2	okie dokie.		
19:28:46.4 INT-2	Potomac in three.		
19:28:49.5 INT-1	roger.		
19:28:52.5 INT-2	Culpeper in two.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:29:05.1 INT-1	whenever you're ready ma'am.		
19:29:06.4 INT-2	okay let's do it.		
19:29:10.3 INT-2	all right.		
19:29:11.9 INT-1	parking brake is released.		
19:29:13.3 INT-2	released.		
19:29:17.9 INT-2	okay before takeoff power control levers are in fly we're gonna leave ourselves in auto we've got twenty two in the mains. transponder's still transponding we've got... approach and uh... Culpeper.		
19:29:35.9 INT-2	crew pax mission equipment secure?		
19:29:38.3 INT-3	left rear secure.		
19:29:39.0 INT-1	right front.		
19:29:39.9 INT-2	okay left front too.		
19:29:41.0 INT-1	coming up.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:29:41.4 INT-2	alright. clear to come up.		
19:29:42.2 INT-3	clear left.		
19:29:52.0 INT-1	ten feet and on the go.		
19:29:54.0 INT-2	you're good to go.		
19:29:55.7 INT-2	and since we're outside the SFRA if you want to you can uh like turn off that landing light but I don't know if you prefer it.		
19:30:00.8 INT-1	nah I wouldn't say I prefer it but I'll just leave it on at this point. we're going to be going back in and I know I will end up forgetting it.		
19:30:07.8 INT-2	totally fair.		
19:30:19.1 INT-1	and clear around right you said to a zero two zero?		
19:30:21.9 INT-2	yup zero two zero. Warrenton.		
19:30:26.4 INT-2	that should keep us out of Culpeper's airspace.		
19:30:41.0 INT-1	alright. zero two zero.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:30:43.0 INT-2	yup.		
19:30:44.3 INT-1	is Culpeper gonna be to our gonna stay to our northwest?		
19:30:48.3 INT-2	uh yes correct.		
19:30:50.6 INT-1	roger.		
19:30:53.6 INT-2	so we're gonna- I tried to keep us away from like centerline.		
19:31:05.5 INT-1	we're coming up to what altitude?		
19:31:09.0 INT-2	thank you for the reminder we are good to climb to...		
19:31:17.8 INT-2	...two thousand feet.		
19:31:19.4 INT-1	come up two thousand.		
19:31:23.6 INT-2	I'm gonna want you to climb to twenty five hundred. elevation does increase a little bit up there.		
19:31:29.4 INT-1	roger.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:31:41.0 INT-2	okay we're gonna cross over the intersection that's our R-P.		
19:31:45.1 INT-2	*		
19:31:46.3 INT-2	and then we can keep going *		
19:31:49.4 INT-1	left on course.		
19:31:59.6 INT-2	just a hair fast but lookin' good.		
19:32:02.0 INT-1	roger.		
19:32:04.2 INT-1	I did not start my clock looks like you did. * * your side. master caution.		
19:32:06.9 INT-2	I did.		
19:32:09.9 INT-2	for the aux fuel.		
19:32:11.1 INT-1	aux fuel roger. we'll just keep a monitor on it make sure we're not out of balance.		
19:32:16.2 INT-1	fifteen hundred for two thousand.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:32:19.7 INT-2	perfect.		
19:32:42.9 INT-1	what's the shelf out here or are we outside the shelf?		
19:32:46.0 INT-2	we're outside the shelf. I believe its forty five hundred bottom but I'll double check.		
19:32:50.9 INT-1	master caution. aux fuel.		
19:32:51.8 INT-2	for the aux fuel thank you.		
19:32:57.4 INT-2	and now that we're high enough I'll try to get flight following again with uh Potomac.		
19:33:01.3 INT-1	roger.		
19:33:04.4 INT-2	we're outside the bravo shelf but the bottom is forty five so we're good.		
19:33:34.0 INT-2	there's Culpeper off the left.		
19:33:43.2 INT-2	* *.		
19:33:54.2 INT-2	the controller sounds a little busy but I'll try him first.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
		19:34:18.9 RDO-2	Potomac Approach PAT two five is a U-H-sixty off of Culpeper requesting V-F-R flight following to Montgomery County.
19:34:43.3 INT-2	let me try a different frequency I think.		
19:34:46.9 INT-2	twenty seven point three two five yeah.		
19:35:21.1 INT-1	um what airport is that out the right?		
19:35:24.6 INT-2	right?		
19:35:28.5 INT-2	I genuinely can't remember right now oh that's um Warrenton yup.		
19:35:31.2 INT-1	that's Warrenton roger.		
19:35:35.3 INT-2	KHWY. [spoken as 'kay highway]		
19:35:39.8 INT-2	we'll be outside of their five mile bubble they should be class G right now.		
		19:35:46.5 RDO-2	Potomac Approach PAT two five a U-H-sixty off of Culpeper requesting V-F-R flight following to Montgomery County.

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
		19:35:53.8 APR-P	standby.
		19:36:09.6 APR-P	PAT two five Potomac Approach IDENT * * for flight following? [transmission broken]
19:36:14.4 INT-2	looking good the intersection that we wanted to cross is right- is coming up.		
19:36:21.0 INT-1	roger.		
19:36:21.9 INT-2	kind of that bright spot below us like at eleven thirty.		
19:36:26.4 INT-1	roger.		
19:36:27.7 INT-2	check our timing.		
19:36:33.5 INT-2	we wanted to cross it at seven twenty three so we're pretty good.		
		19:36:40.0 APR-P	PAT two five Potomac approach IDENT where you looking for flight following to? [transmission broken]
		19:36:46.2 RDO-2	PAT two five IDENT flight following to Montgomery County K-G-A-I.

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
		19:37:05.0 APR-P	PAT two five your radar contact four miles southeast of the Casanova V-O-R at two thousand two hundred maintain * * bravo * altimeter two niner eight seven. [transmission broken]
19:37:16.0 INT-2	couldn't hear what he said two nine something seven.		
19:37:18.2 INT-1	two niner eight seven.		
		19:37:20.0 RDO-2	two niner eight seven PAT two five will remain clear of bravo.
19:37:27.1 INT-2	cool. the lowest shelf we're gonna kind of encounter is two thousand five hundred but we'll be at two thousand at that juncture.		
19:37:35.1 INT-1	roger.		
19:37:38.0 INT-2	slight change to the heading it's like a zero two two now.		
19:37:43.1 INT-1	roger coming right.		
19:38:22.1 INT-1	what is that frequency in two is that Warrenton or Culpeper?		
19:38:25.3 INT-2	that was Culpeper.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:38:27.9 INT-1	roger.	19:38:29.4 APR-P	PAT two five contact Potomac Approach one two five point zero five.
		19:38:34.0 RDO-2	one two five zero five PAT two five.
		19:38:53.4 RDO-2	Potomac Approach PAT two five with you at two thousand.
		19:38:57.0 APR-P	PAT two five Potomac maintain V-F-R altimeter is two nine eight seven.
		19:39:00.7 RDO-2	two nine eight seven PAT two five.
19:39:08.6 INT-2	a little turbulent.		
19:39:10.1 INT-1	little bump.		
19:39:17.7 INT-1	twenty two ten on fuel it's still transferring its good.		
19:39:21.8 INT-2	sweet. you want me to turn it off?		
19:39:23.4 INT-1	nope you're good I was just acknowledging.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:39:42.7 INT-2	we're coming up on another intersection and turning to...		
19:39:49.5 INT-2	...a new heading.		
19:40:00.8 INT-2	* you can turn to a three five two heading.		
19:40:04.6 INT-1	rog-o I'll do that.		
19:40:06.5 INT-3	clear left.		
19:40:56.5 INT-2	all good with the descent and slow down?		
19:40:58.7 INT-1	say again?		
19:40:59.4 INT-2	are we just descending and slowing a little bit all good?		
19:41:01.5 INT-1	*.		
19:41:05.0 INT-2	I'm going to- uh you've got the flight controls?		
19:41:07.8 INT-1	yeah.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:41:08.5 INT-2	okay.		
19:41:09.2 INT-1	what's up?		
19:41:10.3 INT-2	I got the flight controls.		
19:41:11.6 INT-1	yup you got flight controls.		
19:41:12.4 INT-2	I got flight controls.		
19:41:13.4 INT-1	you do have flight controls.		
19:41:17.2 INT-1	sorry about that.		
19:41:18.5 INT-2	what's that?		
19:41:19.3 INT-1	sorry about that.		
19:41:19.7 INT-2	oh yeah it happens. [sound of laugh]		
19:41:22.9 INT-2	climbing back up to two thousand.		
19:41:24.8 INT-1	I've got you at twenty six hundred.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:41:27.7 INT-2	thank you.		
19:41:30.3 INT-2	twenty six hundred I'm sorry I don't follow you.		
19:41:31.6 INT-1	er sorry sixteen hundred.		
19:41:32.5 INT-2	okay yeah.		
19:42:14.0 INT-1	is this the city of Warrenton?		
19:42:15.6 INT-2	yes it is.		
19:42:17.8 INT-2	sweet.		
19:42:21.2 INT-2	strong headwind right now.		
19:42:23.4 INT-1	what's up?		
19:42:23.9 INT-2	uh pretty strong headwind um like one ten indicated.		
19:42:27.4 INT-1	oh yeah its kickin'.		
19:42:28.1 INT-2	ninety three.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:42:29.3 INT-2	'kay.	19:42:30.2 APR-P	PAT two five do you want to take uh the long way around or a shortcut maybe towards Leesburg?
19:42:35.9 INT-2	you got that.		
19:42:36.8 INT-1	you're talking on three?		
19:42:37.7 INT-2	uh yeah three.		
		19:42:38.5 RDO-1	PAT two five we'll uh stay to the west of Leesburg and we'll take the long way around.
		19:42:43.0 APR-P	okay.
		19:42:43.8 RDO-1	thank you for offering.
19:42:45.0 INT-2	oh that's choppy.		
19:42:58.0 INT-1	alright. I'll take flight controls back.		
19:42:59.3 INT-2	okay. you have flight controls.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:43:00.5 INT-1	I have flight controls.		
19:43:00.8 INT-2	you got 'em.		
19:43:01.9 INT-1	two thousand and one ten. seventy nine over the ground.		
19:43:05.5 INT-2	perfect thank you.		
19:43:08.2 INT-1	where do you want me going?		
19:43:09.9 INT-2	* double checking *.		
19:43:13.1 INT-2	* we're going to be turning to the north.		
19:43:16.0 INT-1	clear right? clear right coming right.		
19:43:18.5 INT-2	* * *.		
19:43:25.8 INT-2	three one heading. zero three one.		
19:43:28.2 INT-1	zero three one roger.		
19:43:48.3 INT-1	slowed down a little bit correcting.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:43:51.4 INT-2	good		
19:43:54.4 INT-2	got the wind coming from the left now.		
19:43:56.9 INT-1	yup.		
19:44:30.2 INT-1	so if you were with a P-I what would you kind of recommend to them to counteract the turbulence?		
19:44:35.6 INT-2	fly a little faster so that one fifteen.		
19:44:38.4 INT-1	um yeah what else can they do?		
19:44:43.2 INT-2	well we know it's supposed to be like lower to the ground. we could fly a little higher but still avoid the bravo.		
19:44:52.0 INT-1	anything just like where we're at now without speedin' up? anything they should be thinking about?		
19:44:58.9 INT-2	not overcontrolling the aircraft.		
19:45:00.8 INT-1	yeah. so that comes into power settings right?		
19:45:03.5 INT-2	sure. * *		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:45:05.4 INT-1	so just you know hey. what's giving us what we want. fly that power setting and don't so much worry about the V-S-I and everything else.		
19:45:13.2 INT-2	yeah.		
19:45:15.1 INT-1	coming right to three...		
19:45:16.9 INT-2	if you want it we can set it.		
19:45:19.9 INT-2	uh. set a. uh.		
19:45:28.3 INT-2	friction point essentially.		
19:45:30.8 INT-1	yup.		
19:45:56.3 INT-1	how much further on this leg?		
19:46:00.0 INT-2	double check.		
19:46:10.1 FPS	[sound of rapid beeping similar to stabilator auto mode fail master caution]		
19:46:11.8 INT-1	master caution.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:46:14.1 INT-2	stabilator.		
19:46:14.2 INT-1	you reset that.		
19:46:15.4 INT-1	aircraft is stable.		
19:46:15.9 INT-2	okay. alright.		
19:46:17.0 INT-1	rotor is *.		
19:46:18.9 INT-1	airspeed looks good.		
19:46:20.0 INT-2	see any dip in the nose? alright I'm going to auto control reset one time.		
19:46:24.9 INT-1	roger.		
19:46:27.3 INT-2	and then I need you back to the left please.		
19:46:31.2 INT-1	coming to the left.		
19:46:32.1 INT-3	clear left.		
19:46:32.6 INT-2	* *		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:46:37.2 INT-2	three five two heading.		
19:46:39.9 INT-1	yup. I'm a little high as well. correcting.		
19:46:42.2 INT-1	you said three five two?		
19:46:44.0 INT-2	yes please.		
19:46:45.1 INT-1	rog-o.		
19:46:50.1 INT-1	I thought we went over to the three one heading or whatever.		
19:46:54.9 INT-2	no you're good and then yeah correct down to two thousand a little bit.		
19:46:59.1 INT-1	coming down.		
19:47:08.9 INT-1	continuing left three five two.		
19:47:10.2 INT-3	clear left.		
19:47:11.7 INT-2	thank you.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:47:14.5 INT-3	can I have the heat switched on again for a little bit?		
19:47:17.5 INT-1	say again?		
19:47:18.3 INT-3	can I get the heat on for a little bit?		
19:47:20.0 INT-1	yup.		
19:47:21.1 INT-1	did turn that rheostat down so * turn it back up.		
19:47:26.9 INT-2	I didn't turn it off.		
19:47:28.8 INT-1	no I turned the rheostat down you'll have to turn it back up.		
19:47:37.0 INT-2	to the right? I'm not sure *.		
19:47:38.6 INT-3	yeah turn it to the right.		
19:47:40.6 INT-2	* push it too hard it wouldn't really move much.		
19:47:45.3 INT-1	there it goes wide open.		
19:47:46.5 INT-3	there it goes.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:47:48.1 INT-2	perfect.		
19:47:49.2 INT-3	thank you.		
19:47:49.9 INT-1	yup.		
19:48:05.8 INT-2	and a little bit of wind * you'll come a little bit further to three like four zero we'll try that.		
19:48:10.1 INT-1	clear left.		
19:48:11.3 INT-3	clear left.		
19:48:16.9 INT-2	and come down two thousand.		
19:48:18.7 INT-1	down to two thousand.		
19:48:23.3 INT-1	three four zero rollin' out.		
19:48:26.9 INT-2	good.		
19:48:29.2 INT-2	um okay. the uh we're able to auto control reset with the stabilator but I'm gonna look at the E-P.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:48:36.6 INT-2	see if we got anything. um. but. * it went back into the mode so that's good. if it didn't we... um we'd above forty knots we'd just slew the stabilator to zero and try to fly without.		
19:48:49.9 INT-1	roger. that was at zero. I did not slew it. coming right on course.		
19:48:56.9 INT-2	alright. outside.		
19:48:58.9 INT-1	roger outside.		
19:49:18.3 INT-2	okay we got the beeping warning and the stabilator left auto mode. um * * stabilator slew up switch as necessary. * * * but we didn't notice any nose so we didn't have to arrest that descent. uh we were able to push the auto control once after establishing our comfortable airspeed and thankfully we did regain automatic control so we did not need to position it below the placard airspeed. alright.		
19:49:45.6 INT-1	so how's that supposed to look ma'am?		
19:49:48.2 INT-2	that- E-P?		
19:49:50.7 INT-1	yup. any E-P.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:49:53.0 INT-2	uh more of like just running through FADEC F more deliberately.		
19:49:56.9 INT-1	yup. so uh...		
19:49:58.5 INT-2	it resolved pretty quickly but we should have I should have * more.		
19:50:03.1 INT-1	yup. so I mean that's the whole point of the F-R-C cards right...		
19:50:05.7 INT-2	uh huh.		
19:50:06.2 INT-1	...is to get us away from doing those things automatically.		
19:50:09.1 INT-2	sure.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:50:09.8 INT-1	um yeah we all just pretty much confirmed that it probably is just going to be an auto control reset. but the whole point of the F-R-Cs is 'cause we end up killing ourselves because we do something without confirming and verifying. and so I mean that's yeah the idea is if its not an immediate step we're getting in the checklist we're working as a crew um to ensure we're doing the right thing. again it is a mentality it is a tough mentality to just slow down. * I know how to fix this but I'm going to read the book anyways. but that-that's you know that's what they want. that's what we're looking for so.		
19:50:45.5 INT-2	okay thank you yeah you're absolutely right. i sometimes forget that.		
19:51:03.6 INT-2	we should be coming up to a small private pad just off the right door.		
19:51:08.6 INT-1	roger.		
19:51:11.1 INT-2	and the little town of Middleburg.		
19:51:14.8 INT-2	turn to the right a little bit.		
19:51:17.3 INT-1	coming right. I do see what looks like a small little town over there to the uh one o'clock maybe.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:51:22.5 INT-2	I think so.		
19:51:46.4 INT-1	and there's the little track. alright you heading is zero three two.		
19:51:51.4 INT-1	right turn zero three two.		
19:52:45.3 INT-2	back to the left a little bit.		
19:52:52.1 INT-2	back to the left like uh *.		
19:52:55.8 INT-1	three six zero?		
19:52:57.2 INT-2	yeah.		
19:52:57.5 INT-1	okay coming left.		
19:53:00.2 INT-3	clear left.		
19:53:00.7 INT-2	blown off course.		
19:53:07.9 INT-1	three six zero.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:53:18.7 INT-2	I think it's still pushing us a little bit so a little farther so like three five zero.		
19:53:23.1 INT-1	three five zero.		
19:53:24.6 INT-3	clear left.		
19:53:29.0 INT-1	three five zero.		
19:53:39.5 INT-2	I don't know about you but I'm feeling a little dizzy like looking down and around with the winds or whatever it is but not too bad.		
19:53:44.4 INT-1	roger.		
19:54:02.0 INT-1	it is definitely windy.		
19:54:04.4 INT-2	yeah.		
19:54:12.1 INT-1	but a beautiful night other than that.		
19:54:15.4 INT-2	that's true.		
19:55:24.6 INT-1	Muscle eight where the hell's he at? [said in response to radio traffic, not transcribed.]		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:55:31.3 INT-2	looking for that *.		
19:55:43.5 INT-2	okay crossing * for * point. we're about forty five seconds slow.		
19:55:51.6 INT-1	rog-o.		
19:55:53.0 INT-2	we can pick up a little more speed.		
19:55:54.4 INT-1	yup.		
19:56:09.0 INT-2	alright you can start that turn now. and this is actually to the number one needle.		
19:56:14.1 INT-1	alright turning to number one. clear right.		
19:56:16.3 INT-2	we can use that to our advantage you can set up the nav mode instead of heading if you like.		
19:56:20.2 INT-1	roger got the heading up.		
19:56:21.4 INT-2	* line.		
19:56:23.8 INT-1	master caution aux fuel. resetting.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:56:37.5 INT-2	is that just turbulence or do you need assistance?		
19:56:40.1 INT-1	no that's just turb. I'm trying to turn with uh- yeah we're just gettin' pushed pretty far out of trim. trying to keep her in trim and get that turn in she did not like it. she did not like it.		
19:56:40.8 INT-2	'kay.		
19:56:58.6 INT-2	*.		
19:57:01.7 INT-1	you just said turn to the number one?		
19:57:03.5 INT-2	yes please.		
19:57:05.3 INT-1	tryin' to get that turn in. every time I'm turning right she's...		
19:57:08.5 INT-2	It's gonna blow you there so you can roll out on a heading of like...zero three zero if you want to.		
19:57:12.3 INT-1	yup. roger. rolling out.		
19:57:39.2 INT-2	master caution aux fuel.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
19:57:48.8 INT-1	alright. coming left on course.		
19:57:50.9 INT-2	'kay. we're entering back into the SFRA fairly soon.		
19:57:54.4 INT-1	rog-o. coming back to the right.		
19:58:03.3 INT-1	little high correcting.		
19:58:04.8 INT-2	that's fine. as long as we're below twenty five hundred we're okay.		
19:58:36.5 INT-1	coming left.		
19:58:37.8 INT-2	clear left.		
19:58:39.8 INT-3	clear left.		
19:59:54.8 INT-1	and left on course.		
		19:59:55.8 APR-P	PAT two five traffic at your ten to eleven o'clock and two miles eastbound altitude indicates one thousand eight hundred it's a helicopter.
19:59:56.7 INT-2	*.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:00:05.3 INT-1	that was for you. two miles eastbound.		
		20:00:11.0 APR-P	PAT two five if you hear Potomac acknowledge with an IDENT. traffic at your nine to ten o'clock in two miles eastbound one thousand eight hundred indicated its a helicopter.
20:00:11.2 INT-2	do you see him?		
20:00:12.1 INT-1	nope.		
20:00:20.2 INT-2	do you see him?		
20:00:21.7 INT-1	no. nine to ten o'clock. *.		
		20:00:22.7 RDO-2	* * traffic.
20:00:25.5 INT-1	yeah I got it. tally. coming left.		
20:00:29.1 INT-2	* *.		
20:00:43.2 INT-1	alright you want me to keep chasing this number one needle or-		
20:00:45.0 INT-2	yeah. just avoid the traffic at this point.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:00:46.6 INT-1	yup. I got the traffic out the right door *.	20:00:48.4 RDO-2	PAT two five has the traffic in sight maintaining visual separation. [transmission stepped on by other radio taffic]
		20:00:56.2 RDO-2	PAT two five has traffic in sight. avoiding.
20:00:59.6 INT-2	I hit IDENT when he asked me.		
20:01:01.9 INT-1	say again?		
20:01:06.9 INT-2	okay. you can turn back to the right please.		
20:01:08.9 INT-1	alright coming right behind the muscle twelve aircraft have him in sight.	20:01:18.1 APR-P	PAT two five traffic is no factor.
		20:01:21.2 RDO-2	PAT two five.
20:01:23.6 INT-2	alright back to the right.		
20:01:25.8 INT-1	coming right.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:01:33.0 INT-1	do still have him out the right door.		
20:01:35.7 INT-2	*.		
20:01:36.6 INT-1	continuing around to the right.		
20:01:40.4 INT-2	yeah * lost in the ground lights thanks for the tip.		
20:01:55.7 INT-2	would you be up for me flying for a little while?		
20:01:59.5 INT-1	yup. by all means ma'am.		
20:02:01.0 INT-2	I've got the flight controls.		
20:02:02.3 INT-1	you've got flight controls just know the wind kickin' out of the left definitely uh want to keep it in trim.		
20:02:07.0 INT-2	I can feel that.		
20:02:11.0 INT-1	you have flight controls.		
20:02:12.3 INT-2	I've got 'em thanks.		
20:02:28.5 INT-1	still got the Muscle aircraft out the right door.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:02:30.0 INT-2	thank you.		
20:02:32.2 INT-1	two hundred feet.		
		20:02:32.9 APR-P	PAT two five you can uh proceed on course to Gaithersburg inside the S-F-R-A and outside the class bravo airspace.
		20:02:38.7 RDO-1	just outside the class bravo on course to Gaithersburg PAT two five.
20:02:43.4 INT-2	okay correcting my altitude.		
20:02:48.1 INT-2	a little fast on the groundspeed * headwind.		
20:02:53.1 INT-2	tailwind.		
20:03:02.4 INT-1	alright what's the next...		
20:03:05.8 INT-2	*.		
20:03:06.8 INT-1	nineteen miles. cool.		
20:03:08.6 INT-2	yeah we got that. I've got a couple...* for the T-D-H but that one will work too.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:03:15.1 INT-1	alright you want me to do anything or just leave it at R-P two?		
20:03:17.1 INT-2	um...yeah leave it unless you want to pull your iPad out use naving or use the T-D-H I can try to get you we got on course just fine.		
20:03:25.1 INT-1	okay. yup. you're good.		
20:03:32.8 INT-2	but we were just coming up to the firehouse so timing is good so far.		
20:03:36.2 INT-1	alright I got uh the river I got Muscle eight er Muscle twelve out the right door one o'clock he's gonna pick up the river looking for Leesburg international airport uh don't quite have it in sight.		
20:03:47.9 INT-2	that's an international airport I had no idea.		
20:03:49.9 INT-1	no I doubt it. I don't think they are.		
		20:03:52.3 APR-P	PAT two five just confirm you're landing at Gaithersburg is that correct?
		20:03:54.7 RDO-1	* * PAT two five we're gonna be landing about ni— five miles to the northeast at a uh helipad there in Olney Maryland so five miles to the northeast of Gaithersburg.

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:03:54.8 INT-2	I want to make sure we don't-		
20:04:11.9 INT-2	want to make sure we're not getting too close to the fifteen for the shelf before we reenter the SFRA.		
20:04:17.0 INT-1	*.		
20:04:19.6 INT-2	let me check my iPad.		
		20:04:20.4 APR-P	PAT two five go ahead again please.
		20:04:22.9 RDO-1	negative uh do request entry into the S-F-R-A we will not be landing at Gaithersburg um landing five miles to the north.
		20:04:29.0 APR-P	five miles five miles north * *.
20:04:46.7 INT-1	yup you're good here.		
20:04:48.0 INT-2	thank you.		
20:04:54.7 INT-1	there's one two three zero seven five per *.		
20:04:58.1 INT-1	Muscle aircraft is headed down the river negative factor for us.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
		20:05:26.7 APR-P	PAT two five contact Potomac Approach one two six point one twenty six one.
		20:05:31.0 RDO-1	one twenty six one have a good night PAT two five.
20:05:35.1 INT-1	right front's inside.		
20:05:37.0 INT-2	got your scan.		
		20:05:59.0 RDO-1	approach PAT two five's with you two thousand.
20:06:04.1 INT-2	back outside.		
20:06:05.5 INT-1	back outside on front right.		
20:06:25.7 INT-1	alright you warm enough for a bit @[CrewChief]?		
20:06:27.8 INT-3	yup.		
20:06:28.3 INT-1	all right.		
20:06:34.9 INT-2	don't like the heater?		
20:06:36.8 INT-1	oh I do I'm just pretty warm.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:06:38.6 INT-2	yeah.		
20:06:40.8 INT-1	it's warm.		
		20:06:43.0 RDO-1	Potomac Approach PAT two five with you two thousand.
20:06:54.8 INT-1	or not. right front's inside.		
20:06:57.9 INT-2	I got your scan.		
20:07:17.4 INT-2	eighty indicated but one oh eight ground that's wild.		
20:07:21.9 INT-1	gettin' it.		
		20:07:39.5 RDO-1	approach PAT two five level with you two thousand two hundred.
20:07:47.8 INT-1	oh that's why. mother trucker.		
20:07:50.8 INT-2	oh wrong radio?		
20:07:54.7 INT-1	I couldn't see the last little frequency.		
20:07:58.5 INT-1	left front's inside. right front's inside.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:08:00.2 INT-2	'kay.		
		20:08:30.6 APR-P	PAT two five Potomac.
		20:08:33.9 RDO-1	my apologies sir PAT two five is with you twenty one hundred feet.
		20:08:38.4 APR-P	roger that uh maintain V-F-R outside of bravo and uh you going to Gaithersburg or outside of Gaithersburg there?
		20:08:44.0 RDO-1	we'll be just outside of Gaithersburg about uh five miles to the northeast PAT two five outside the bravo.
		20:08:48.7 APR-P	roger that. and just let me know when you have the landing area in sight.
		20:08:51.9 RDO-1	two five wilco.
20:09:22.8 INT-1	alright still got about a thousand pounds over in the externals. and twenty one twenty auto mode still working good.		
20:09:31.8 INT-2	'kay.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:09:33.1 INT-1	so got about an hour and twenty minute flight almost an hour thirty now hour and a half looking good on time.		
20:09:42.9 INT-2	shouldn't need to go to the FARP.		
20:09:45.7 INT-1	say what?		
20:09:47.2 INT-2	unless you wanted to.		
20:09:49.2 INT-1	unless you want to do what?		
20:09:50.3 INT-2	we shouldn't need to use the FARP unless you want to.		
20:09:52.8 INT-1	uh yeah no. uh well I don't know. when we burn this off it'll be close.		
20:09:58.4 INT-1	might as well use 'em I hate not using 'em. I always feel really bad I'm like aw man they gotta sit out there.		
20:10:00.0 INT-2	yeah. yeah let's do it.		
20:10:04.9 INT-3	I mean it's kind of their only job so...		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:10:07.2 INT-1	I know so it's like when you don't use 'em. I don't know I always feel bad.		
20:10:12.5 INT-3	I never feel bad about it. they barely do their job right anyway.		
20:10:15.6 INT-1	[sound of laugh] better watch it. you got the O-I-C of the FARP on board.		
20:10:18.7 INT-2	[sound of quick laugh] no if you got some issues with them let me know.		
20:10:23.9 INT-3	haven't had issues with the new group yet but the last group definitely had some friction.		
20:10:28.2 INT-2	I remember like that issue when like they made the trainee stand far away.		
20:10:33.5 INT-3	yeah.		
20:10:34.2 INT-2	were there other issues?		
20:10:35.5 INT-3	uh. a repeat issue with that and then um...there was something else but I can't remember off the top of my head.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:10:48.7 INT-2	I do feel for them though they only got like seven people here. and realistically we need like four on at a time at the FARP and so we don't even get that.		
		20:11:04.7 RDO-1	Montgomery County traffic PAT two five single U-H sixty. twenty three hundred feet we'll be crossing uh from the west to the east about five miles to the west at this time Montgomery County.
		20:11:22.4 RDO	[sound similar to eight mic clicks, consistent with operating pilot controlled lighting (PCL)]
20:11:26.1 INT-1	boom coolest thing you can do as a pilot.		
20:11:28.2 INT-2	yeah that light you're right that is really cool.		
20:11:31.3 INT-1	I just turned lights on from two thousand feet.		
20:11:33.7 INT-2	[sound of laugh]		
20:11:35.1 INT-1	what kind of crazy world are we living in?		
20:11:37.4 INT-2	that is pretty wild.		
20:11:38.8 INT-1	Thomas Edison would have never imagined.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:11:40.9 INT-2	[sound of laugh]		
20:11:48.1 INT-1	alright I'll...you're three point nine from your R-P. definitely think we're going to have to land to the north.		
20:11:53.8 INT-2	okay.		
20:11:59.5 INT-2	yeah.		
20:12:00.5 INT-1	this wind is uh...		
20:12:01.9 INT-2	quite a headwind.		
20:12:03.0 INT-1	...not gonna make it super safe to be landing to the south.		
20:12:07.6 INT-2	yeah.		
		20:12:27.0 RDO-1	Montgomery County traffic. PAT two five single U-H sixty two thousand feet. three miles to the west. * transitioning west to east. Montgomery County.
20:12:41.4 INT-2	you err on the side of caution with calling more C-TAFs *.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:12:45.2 INT-1	I do. it doesn't cost me a thing in the world. and it may save my bacon one day when somebody's walk- flying around not talking to approach or whatever.		
20:12:53.7 INT-2	yeah.		
20:12:54.3 INT-1	so I- I use 'em pretty often. I don't have any reason not to. and we are within two miles of Gaithersburg right now.		
20:13:01.0 INT-2	yeah. we're close to this one.		
20:13:15.7 INT-1	alright. you're one point four to your R-P.		
20:13:20.1 INT-2	okay. slow down come down a little bit.		
20:13:42.3 INT-2	wow that's wild. I'm slowing down and I'm still at a hundred knots ground.		
20:14:01.9 INT-1	point seven. what was your R-P?		
20:14:04.4 INT-2	what about it?		
20:14:05.5 INT-1	what was it?		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:14:06.0 INT-2	um...I'm sorry what do you mean what was it?		
20:14:09.1 INT-1	alright what am I looking for on the ground?		
20:14:10.6 INT-2	um. I can't remember I have to look at my notes.		
20:14:12.9 INT-1	yeah no worries. just use the needle.		
20:14:15.7 INT-2	that'll work. and then eighty six should be the uh Olney.		
20:14:23.7 INT-1	alright two point two miles per the number one needle.		
20:14:26.8 INT-2	'kay. we'll cross over it to the east and we'll do a low pass.		
20:14:33.3 INT-1	alright. we are locked released crew passengers mission equipment.		
20:14:36.2 INT-2	*.		
20:14:36.4 INT-3	crew secure.		
20:14:37.6 INT-1	right front.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:14:39.7 INT-1	alright looks like I already have the pad. I don't know what that is don't get sucked into that.		
20:14:44.2 INT-2	I won't. okay. it's an intersection that we're supposed to cross over *.		
20:14:51.4 INT-1	two K out. two point eight.		
20:14:56.6 INT-1	alright don't get too slow.		
20:14:57.9 INT-2	okay.		
20:15:00.0 INT-1	again this is one of those you know. * talk about L-T-E. where you get slow. start pulling in that power.		
20:15:07.2 INT-2	yeah *. okay.		
20:15:11.1 INT-2	yeah we got the wind coming of the...left door there.		
20:15:14.1 INT-1	just off the left door yeah.		
20:15:17.5 INT-1	alright we are- you can start coming down though.		
20:15:19.7 INT-2	'kay. coming down some more. clear obstacles.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:15:23.4 INT-1	and I'm thinking that's the pad out there at the left uh it's gonna be coming to the eleven to the ten right.		
20:15:31.8 INT-3	I'm searching for it. *.		
20:15:33.5 INT-1	I can't get sucked in by that damn light.		
20:15:36.6 INT-2	that light's really bright.		
20:15:39.5 INT-2	that's why. * what the M-S-L is for it.		
20:15:43.9 INT-1	yup so there's the pad down there.		
20:15:44.7 INT-2	yeah I see it.		
20:15:46.1 INT-1	probably just make a-		
20:15:47.5 INT-2	make a little left traffic.		
20:15:48.5 INT-1	left or right probably right cause if we're gonna land to the north.		
20:15:52.1 INT-2	oh that's a good point. okay. yeah.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:15:56.5 INT-1	got the pad @[Crew Chief]?		
20:15:57.4 INT-3	yes I do.		
20:15:58.3 INT-1	alright roger.		
20:15:59.8 INT-2	okay so I'm going to turn right here. okay.		
20:16:03.3 INT-1	keep that airspeed up so now * low.		
20:16:12.0 INT-2	what's my ground speed? I feel like I'm kind of moving but I see my indicated is like thirty.		
20:16:16.7 INT-1	ground speed is sixty.		
20:16:18.9 INT-2	yeah.		
		20:16:21.9 RDO-1	and PAT two five is landing assured we'll call you on the go.
20:16:35.3 INT-2	clear right.		
20:16:37.2 INT-1	roger right. a little right pedal.		
20:16:40.7 INT-2	* trim.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
		20:16:44.4 RDO-1	approach PAT two five is landing assured we'll call you on the go.
20:16:48.6 INT-2	yeah. I don't have the pad in sight quite yet.		
20:16:51.4 INT-1	no you're all good.		
20:16:54.7 INT-1	one point five K out. gonna be out to your right so look at your number one needle.		
20:16:59.6 INT-2	yup I see it. gonna turn towards it.		
20:17:05.4 INT-2	there's the * trees I'm looking for got it.		
20:17:09.1 INT-1	start your slow back start your approach.		
20:17:10.8 INT-2	yeah slowing back.		
20:17:12.8 INT-2	another set of trees I can't quite see it but I know where it is.		
20:17:25.1 INT-1	before landing is good.		
20:17:26.7 INT-2	thank you.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:17:31.4 INT-1	got a left quartering...wind.		
20:17:33.6 INT-2	yeah I feel it.		
20:17:39.8 INT-1	remember there is a fence on the north side of this pad.		
20:17:42.2 INT-2	yeah. thank you. I see the sock I believe.		
20:17:46.0 INT-1	yup I have the windsock in sight.		
20:17:50.9 INT-2	probably still coming in a little too fast.		
20:17:55.9 INT-2	not so bad.		
20:18:01.1 INT-2	see that big tree on the right? am I clear of that?		
20:18:04.0 INT-1	you are.		
20:18:06.1 INT-2	yeah.		
20:18:08.0 INT-1	go around go around go around.		
20:18:09.1 INT-2	go around.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:18:12.8 INT-2	turn to the right. * building.		
20:18:17.0 INT-1	gettin' that airspeed up.		
20:18:18.7 INT-2	* airspeed. see the deer on the ground. I don't really see that fence but I know where it should be.		
20:18:22.9 INT-1	roger. climbing away from the hazards.		
20:18:28.6 INT-2	* power to the...		
20:18:31.4 INT-1	are you going- which way are you going ma'am?		
20:18:32.7 INT-2	I was gonna go right sorry do you want me to go left?		
20:18:34.2 INT-1	uh I think it'd beee probably better. because you're turning in to the wind right?		
20:18:36.4 INT-2	okay. yeah.		
20:18:38.8 INT-1	so again turn into that wind. use it. get more lift. the wind's gonna give you all the lift. and not the uh- whereas you're slow you're only indicating like forty knots...		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:18:49.4 INT-2	yeah.		
20:18:50.3 INT-1	if you turn right then you're going to get that booming tailwind.		
20:18:53.3 INT-2	okay good point.		
20:18:53.8 INT-1	so they always want to fly into the wind as much as we can let 'em.		
20:18:57.8 INT-2	okay. it's out the right door now.		
20:19:00.5 INT-1	what's that?		
20:19:01.2 INT-2	uh the winds should be out the- out of your right- off the right door.		
20:19:03.8 INT-1	yup. wind's out the right door. absolutely.		
20:19:10.4 INT-2	yup. alright clear left.		
20:19:15.9 INT-3	clear left.		
20:19:22.5 INT-2	* *.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
		20:19:31.3 RDO-1	approach PAT two five is making pattern work uh we'll call you when we're headed back.
20:19:49.1 INT-2	light's bright.		
20:20:06.6 INT-2	clear to go left.		
20:20:08.0 INT-3	clear left.		
20:20:21.2 INT-2	*. yeah I'm not sure I'm. I'm.		
20:20:25.4 INT-1	what's up?		
20:20:26.2 INT-2	this doesn't feel right.		
20:20:27.8 INT-1	yup. you're good. feels like you're gettin' your # kicked with some uh wind to the left right?		
20:20:31.8 INT-2	yes.		
20:20:32.2 INT-1	so go ahead and start your approach.		
20:20:34.2 INT-1	okay.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:20:36.0 INT-2	no indicated airspeed even though I'm barely pulling the aft cyclic.		
20:20:39.0 INT-1	yup. * pull in some power.		
20:20:40.7 INT-2	okay. pullin' in a little more.		
20:20:42.3 INT-1	should have pad in sight.		
20:20:44.5 INT-2	I do I'm too high again though.		
20:20:46.5 INT-1	nope you're good. so I'd maybe a left pedal a little bit to put yourself into that wind.		
20:20:51.1 INT-2	'kay.		
20:20:54.3 INT-1	continue. you can start your approach. you're gonna be a little high.		
20:20:59.0 INT-2	yeah I- I'm not comfortable landing there.		
20:21:01.0 INT-1	alright cool.		
20:21:02.4 INT-2	I'm clearing out. I probably could but. its handling funny.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:21:08.7 INT-1	alright. mind if I give it a shot?		
20:21:11.0 INT-2	yeah please.		
20:21:11.9 INT-1	alright. I have flight controls.		
20:21:13.3 INT-2	you got 'em.		
20:21:13.6 INT-1	I do.		
20:21:16.6 INT-1	alright yeah so I mean the first thing I want is airspeed. the faster the better she's gonna want to fly.		
20:21:19.3 INT-2	yup.		
20:21:27.3 INT-1	and clear around left.		
20:21:28.4 INT-2	clear left.		
20:21:28.8 INT-3	clear left.		
20:21:32.1 INT-1	and keeping her in trim will be the other big thing.		
20:21:36.1 INT-2	okay.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:21:37.0 INT-1	so the more we can keep her in trim the better it's going to feel it's gonna feel more natural we're not gonna get in that proprioceptive of the seat of our pants flying.		
20:21:48.3 INT-2	'kay.		
20:21:48.9 INT-1	whereas if we can keep her in trim it feels more natur-al.		
20:21:53.5 INT-1	it does feel smoother now than it was.		
20:21:58.8 INT-1	I do not have the L-Z call my turn.		
20:22:03.1 INT-2	no worries.		
20:22:06.1 INT-2	clear * left.		
20:22:07.8 INT-1	clear left.		
20:22:09.2 INT-3	clear left.		
20:22:11.2 INT-1	again I'm lowering that nose. I'm not trying to slow down and I'm trying to keep her in trim.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:22:15.6 INT-2	okay.		
20:22:19.2 INT-1	still don't have the L-Z in sight.		
20:22:20.8 INT-2	that's okay. it's at the end.		
20:22:23.3 INT-3	at uh nine thirty ten o'clock now.		
20:22:24.3 INT-2	* number one.		
20:22:25.5 INT-1	* roger got it in sight.		
20:22:27.8 INT-2	field.		
20:22:33.4 INT-1	and starting my approach. now I'm gonna put her left nose. I know where the wind's coming from. that's gonna give me my best power margins.		
20:22:49.9 INT-1	gonna just keep pushing that nose over. pulling in my power sixty percent.		
20:23:01.8 INT-1	and call me clear of the trees on the left please sir.		
20:23:05.2 INT-3	roger.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:23:21.6 INT-3	continue forward.		
20:23:23.0 INT-1	continuing forward.		
20:23:26.1 INT-3	mains are clear of obstacles.		
20:23:27.7 INT-1	alright coming down.		
20:23:29.7 INT-2	I do not have sight of the L-Z anymore.		
20:23:31.2 INT-1	roger.		
20:23:31.6 INT-3	tail's clear.		
20:23:33.0 INT-1	alright coming down.		
20:23:34.6 INT-1	that's seventy two percent. if anything happens we're going forward. straight forward.		
20:23:41.3 INT-2	good.		
20:23:44.7 INT-3	do you have the pad in sight sir?		
20:23:46.6 INT-1	a-firm sliding right.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:23:48.0 INT-3	roger.		
20:23:50.4 INT-2	got the fence.		
20:23:52.8 INT-3	mains are over the concrete continue forward.		
20:23:54.7 INT-1	continuing forward.		
20:23:57.1 INT-3	continue forward for five...four...three...two...one...tail's over clear on left.		
20:24:07.6 INT-1	oh sorry. a little nose left tail right.		
20:24:10.4 INT-3	uh.		
20:24:11.0 INT-2	clear left. definitely safe left.		
20:24:13.4 INT-1	and coming down.		
20:24:14.6 INT-3	clear left.		
20:24:22.3 INT-2	okay. pressure * brake *.		
20:24:25.1 INT-1	oh just hold 'em?		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:24:26.6 INT-2	you want to hold 'em? ok cool.		
20:24:27.6 INT-1	yup.		
20:24:29.3 INT-1	alright um. that wind definitely makes it tricky.		
20:24:33.0 INT-2	yeah.		
20:24:33.7 INT-1	and its uh...yeah with it blowing obviously the wind sock doesn't look like its blowing very much. the windsock has it kind of...I guess the windsock has it blowing as a three zero zero. so that's the thing with a helipad I mean you can kind of come in any way you want as long as you as long as you can make aware. obviously those lights out the right door and that tower make it uh not conducive to be over here.		
20:24:59.6 INT-2	good challenge though. worth a try.		
20:25:02.7 INT-1	but other than that- if you want to make another pattern in here or whatever you want to do.		
20:25:08.5 INT-2	yeah hones- I uh when we got to the * at that point that's kind of probably like where I was like yeah I don't want to do this cause I can't really see it like maybe I need to be higher in my chair or something but I kind of lost any * flying. yeah I'm good with the demonstration.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:25:25.4 INT-1	alright.		
20:25:28.8 INT-2	but we can uh go on back tooo... go back south unless you want to do another pattern or do anything else.		
20:25:35.3 INT-1	nope. no I'm good.		
20:26:01.9 INT-1	try seventy fifty six maybe. to Chain Bridge if that's where you want to go.		
20:26:07.8 INT-2	* Cabin John * *.		
20:26:42.7 INT-2	make sure I'm not gonna go through anybody's airspace.		
20:26:45.3 INT-1	rog-o.		
20:26:46.9 INT-2	so yeah we're going to maybe need to call to Gaithersburg but if we just go due south yup through route seven to Cabin John we'll be perfect.		
20:26:55.4 INT-1	cool.		
20:26:55.9 INT-2	*.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:26:58.6 INT-2	altitude up to fifteen. * *.		
20:27:07.3 INT-2	and get up with tower pretty soon alright. talk to Potomac again and then go back and switch to tower pretty soon then I can make a call to uh Montgomery.		
20:27:17.7 INT-1	alright.		
20:27:20.0 INT-2	okay. uh go ahead and take off north if you like and then turn south towards the point.		
20:27:26.4 INT-1	alright.		
20:27:30.2 INT-1	before takeoff.		
20:27:32.0 INT-2	okie dokie.		
20:27:36.6 INT-2	alright. that's not the right route.		
20:27:43.6 INT-2	okay. got it. alright before takeoff engine power control levers are at fly we've got twenty one hundred eighty in the mains four hundred each side tank...in our tanks we are * avionics right now. passengers mission equipment secure.		
20:28:03.2 INT-1	right crew.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:28:04.3 INT-3	left crew.		
20:28:05.2 INT-1	alright. coming up.		
20:28:08.9 INT-3	clear left.		
20:28:10.0 INT-1	roger right.		
20:28:17.2 INT-1	and we're above all hazards on the right.		
20:28:20.3 INT-2	and we're clear on the left.		
20:28:31.9 INT-2	tower one o'clock.		
20:28:33.8 INT-1	tally.		
20:28:35.9 INT-1	* * damn look at all them deer.		
20:28:38.7 INT-2	a lot.		
20:28:39.2 INT-1	holy #.		
20:28:41.0 INT-2	are you a hunter?		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:28:42.0 INT-1	I am. clear left.		
20:28:43.5 INT-3	clear left.		
20:28:44.1 INT-2	on the left.		
20:28:44.7 INT-1	I just saw our shadow and almost pooped myself. that was # terrifying.		
20:28:49.2 INT-2	[sound of laugh] oof.		
20:29:02.0 INT-1	you just want me to come all the way to the south?		
20:29:04.3 INT-2	yup that'll do.		
20:29:05.5 INT-1	alright coming up a thousand feet and to the south.		
20:29:07.8 INT-2	perfect. once we get to a thousand I will try to contact approach.		
20:29:15.9 INT-1	how far are we from Montgomery County?		
20:29:20.0 INT-2	double checking...about four miles.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
		20:29:25.6 RDO-1	Montgomery county PAT two five single U-H sixty a thousand feet operating four miles to the east. PAT two five.
20:29:36.7 INT-1	that was a terrible # call.		
20:29:38.0 INT-2	[sound of quick laugh]		
		20:29:41 TWR-A	[All radio transmissions audible in the ICS are transcribed after this time. Radio transmissions from Washington Tower presented in italics occurred simultaneously to other intra-aircraft communication between the crew, while the crew was talking.]
		20:29:41.3 RDO-2	Potomac approach PAT two five is with you again at one thousand feet off of Montgomery requesting V- uh flight following back to Davison.
20:29:53.7 INT-1	I don't think he can hear us you want me to come up or you just want to go up with National?		
20:29:57.3 INT-2	I'll just go straight to National I don't think anyone will be upset. inside real quick.		
20:30:02.0 INT-1	roger.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
		20:30:19.1 RDO-2	Washington Tower PAT two five U-H sixty off of Montgomery requesting flight following back to Davison.
20:30:27.2 INT-2	wow not flight following *.		
20:30:28.1 INT-1	ha.		
20:30:33.7 INT-2	hopefully they didn't hear that but they probably did.[sound of quick laugh]		
20:30:41.5 INT-1	they just said uh we'll ignore that.		
20:30:43.7 INT-2	yeah. I don't hear any traffic on there though. we don't need to talk to them yet.		
		20:30:49.4 TWR-A	<i>* in sight departing runway three three maintain-maintain visual separation.</i>
20:30:54.7 INT-2	we're below twenty five hundred.		
		20:30:55.9 TWR-A	* * *.
20:30:56.6 INT-1	yup.		
		20:30:58.4 TWR-A	disregard Southwest * * runway runway one.

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
		20:31:05.0 TWR-A	Bluestreak fifty six seventy three contact Potomac.
		20:31:08.1 RDO-?	[sound similar to twelve mic clicks, consistent with operating pilot controlled lighting (PCL)]
20:31:10.9 INT-2	more lights.		
20:31:13.5 INT-1	I was trying to find it. this time for real.		
20:31:20.6 INT-2	* at the back right of the aircraft.		
		20:31:25.4 TWR-A	<i>Skywest * traffic on a two mile final for runway three three no delay runway one clear for takeoff.</i>
20:31:24.9 INT-1	yeah there it is.		
20:31:28.0 INT-1	there's fifteen hundred feet.		
20:31:30.4 INT-2	okay. * altitude. fifteen hundred's good.		
		20:31:33.9 TWR-A	Bluestreak fifty two sixty nine Washington Tower runway one line up and wait traffic landing three three.
20:31:39.2 INT-2	alright number one needle. I'll try them again.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:31:42.2 INT-1	alright coming right on the number one.		
		20:32:00.1 RDO-2	National Tower PAT two five requesting Cabin John route one route four Davison.
		20:32:09.2 TWR-A	PAT two five Washington Tower * two niner eight nine say position. [transmission broken]
		20:32:13.9 RDO-2	two niner eight nine PAT two five proceed as requested.
20:32:18.2 INT-2	alright. is he pretty muffled?		
		20:32:19.0 TWR-A	<i>Skywest fifty eight * contact departure.</i>
20:32:23.7 INT-1	yeah. I definitely didn't catch what he said. I'm glad you did.		
20:32:28.8 INT-2	I don't know if that last one * not. * before that though.		
20:32:38.9 INT-2	huh. volume's at eighteen it's loud enough.		
		20:32:44.4 TWR-A	<i>Bluestreak fifty two sixty nine winds are three one direct one eight gust two six * * takeoff. [transmission broken]</i>

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:32:49.5 INT-1	okay ma'am you can fly. sounds like you've communicated all that needs to be communicated.		
		20:32:53.0 TWR-A	<i>Bluestreak</i> * *.
20:32:54.0 INT-2	okay sounds good. do you want me to fly?		
20:32:55.9 INT-1	sure.		
20:32:56.5 INT-2	alright I have flight controls.		
20:32:57.3 INT-1	you do have flight controls.		
20:32:58.1 INT-2	yup.		
20:33:09.6 INT-1	right front's inside.		
20:33:11.4 INT-2	rog.		
		20:33:16.6 TWR-A	PAT two five Washington Tower IDENT.
		20:33:19.4 RDO-1	there goes PAT two five.
20:33:21.6 INT-2	IDENT.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
		20:33:24.6 TWR-A	* * * squawk * * gust two six runway one. [transmission broken]
		20:33:33.4 TWR-A	PAT * * north of the * say your request. [transmission broken]
		20:33:41.1 RDO-1	PAT two five is looking for Cabin John route one to route four to Davison.
		20:33:46.7 TWR-A	PAT two five approved.
		20:33:48.1 RDO-1	approved two five.
		20:33:49.2 TWR-A	* contact departure.
20:33:51.0 INT-2	*.		
		20:34:03.6 BLJK1	* * zone two * route one heading boardwalk. [transmission broken]
		20:34:13.9 TWR-A	Blackjack one Washington Tower standby.
		20:34:16.6 BLJK1	*.
20:34:17.8 INT-1	right front's back outside.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:34:18.5 INT-2	com sounds really crappy.		
20:34:21.0 INT-2	do they have a U-H-F?		
		20:34:21.7 TWR-A	<i>* winds are three two * * mile final runway one *. [transmission broken]</i>
20:34:24.3 INT-2	we got a master caution for the aux fuel.		
20:34:26.2 INT-1	roger. what'd you say right before that?		
20:34:30.0 INT-2	oh I was gonna do they have a U-H-F...or don't they?		
		20:34:30.7 TWR-A	<i>Bluestreak fifty * left uh left turn at sierra.</i>
20:34:32.3 INT-1	um...		
20:34:35.9 INT-2	he still sounds really crappy.		
20:34:37.3 INT-1	...they do but uh I don't know if they use it.		
		20:34:37.8 TWR-A	<i>Brickyard * * cross runway three three * *. [transmission broken]</i>

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:34:39.2 INT-2	master caution for the aux fuel.		
20:34:42.8 INT-1	roger.		
		20:34:45.9 TWR-A	and Blackjack one you said you want to do zone three zone four?
		20:34:50.1 BLJK1	tower for Blackjack one request zone three zone two split the p's followed by route one * *. [transmission broken]
		20:35:03.9 TWR-A	Blackjack one approved report landing assured.
		20:35:07.0 BLJK1	Blackjack one approved.
20:35:08.1 INT-1	split the p's what?		
		20:35:09.4 TWR-A	<i>Bluestreak fifty twenty five contact departure.</i>
20:35:10.4 INT-2	*.		
20:35:12.3 INT-2	I thought they didn't want us doing that.		
		20:35:13.4 TWR-A	<i>good night. * * *.</i>

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:35:13.9 INT-1	Blackjack out here running amok.		
20:35:17.1 INT-2	[sound of laugh]		
		20:35:19.3 MUSL12	tower Muscle twelve request zone four.
		20:35:24.0 TWR-A	* Washington Tower winds are three two direct one six gust six traffic * arrival runway one *. [transmission broken]
		20:35:34.3 TWR-A	Muscle twelve you said zone four?
		20:35:35.7 MUSL12	a-firm.
		20:35:36.8 TWR-A	Muscle twelve zone four approved.
		20:35:38.4 MUSL12	* * twelve. [transmission broken]
		20:36:10.8 TWR-A	Bluestreak * forty seven Washington Tower winds are three two * * *. [transmission broken]
		20:36:22.3 TWR-A	Bluestreak * * turn left *. [transmission broken]
		20:36:35.8 TWR-A	Brickyard * * hold position runway three three * one more arrival off runway one. [transmission broken]

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:36:41.3 INT-1	just for your awareness I put them in two.		
20:36:44.5 INT-2	okay thanks.		
20:36:46.4 INT-1	see if it'll pick them up a little better.		
20:36:51.2 INT-1	you are five K from Cabin John I'm sure you have it in sight.		
20:36:56.4 INT-2	I believe that's it that bridge.		
20:36:58.4 INT-1	a-firm.		
		20:37:18.9 TWR-A	Blackjack one do you have uh Muscle thirteen just two miles north of you one thousand one hundred feet? [sound of rapid beeping consistent with conflict alert audible in background]
		20:37:26.1 TWR-A	Muscle thirteen traffic just (four) miles south of you Blackjack one * [transmission broken]
		20:37:29.7 MUSL13	Muscle thirteen searching for traffic.
		20:37:31.1 BLJK1	visual sep.
		20:37:32.2 TWR-A	visual separation approved Blackjack one.

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
		20:37:34.0 TWR-A	* fifteen thirty one left turn november ground point seven.
		20:37:39.1 TWR-A	Brickyard fifty seven * winds are three two direct one three gust two six traffic on a two mile final for runway one no delay runway three three clear for takeoff.
		20:37:50.2 TWR-A	and Muscle thirteen do you (see/have) that Blackjack in sight?
		20:37:53.2 MUSL13	negative for Muscle thirteen still searching.
		20:37:55.1 TWR-A	Blackjack correction muscle thirteen traffic south * four miles south no factor * they're * eight hundred. [transmission broken]
		20:38:02.1 MUSL13	<i>Muscle thirteen still searching for traffic we'll be heading northbound * *.</i>
20:38:04.0 INT-1	left seat's inside.		
20:38:05.5 INT-2	say again? you're good I got your scan.		
		20:38:05.6 TWR-A	*.
20:38:11.3 INT-2	thirteen at Cabin John.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
		20:38:14.6 TWR-A	Bluestreak fifty one seventy Washington tower winds are three two direct one three gust two six * *. [transmission broken]
		20:38:27.1 TWR-A	Brickyard fifty seven (fifty/sixty) two contact *.
		20:38:31.8 TWR-A	see ya.
20:38:35.8 INT-1	what's up ma'am where we going?		
20:38:38.1 INT-2	down the river. uh route one to four.		
20:38:42.7 INT-1	uh you just took a right turn which is gonna take us back to Great Falls and *.		
		20:38:46.4 TWR-A	<i>Bluestreak fifty one twenty five * aircraft four mile final runway one line up and wait.</i>
20:38:47.2 INT-2	oh excuse me. okay * I should be headed...alright. coming to the...left.		
20:38:57.4 INT-3	clear left.		
20:38:58.2 INT-1	I mean I'm all game for it but we just gotta let the dude know.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:39:00.8 INT-2	[sound of quick laugh] no totally fair.		
20:39:04.6 INT-1	I am all game for goin to shake up some Great Falls at twenty feet tonight. we didn't brief it so we'll have to make a blood pact.		
20:39:10.5 INT-2	nope. right.		
		20:39:11.4 TWR-A	<i>* four seven two left turn at november one *. [transmission broken]</i>
20:39:14.9 INT-1	alright a little right pedal. yeah that'll help it feel smoother.		
20:39:19.3 INT-2	let's see if it feels smoother.		
		20:39:21.8 TWR-A	<i>* keep your speed up * * right behind you. [transmission broken]</i>
20:39:28.8 INT-2	I'm with it alright. *.		
		20:39:30.3 TWR-A	Bluestreak fifty one seventy two winds are three two direct one two gust two six traffic's on a four mile final without delay runway one clear for takeoff.
		20:39:45.3 MUSL12	tower Muscle twelve request uh direct back to Andrews.

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
		20:39:53.7 TWR-A	Muscle twelve approved.
		20:39:55.4 MUSL12	Muscle twelve. and request freq change.
		20:39:58.2 TWR-A	* stand by for *. [transmission broken]
		20:39:59.9 MUSL12	copy.
20:40:06.2 INT-2	okay.		
20:40:09.0 INT-1	alright down to thirteen hundred feet we're already a little low.		
20:40:11.7 INT-2	alright. oh wow yeah okay.		
20:40:16.8 INT-1	thirteen hundred feet to Chain Bridge at Chain Bridge down to seven hundred of the feets.		
20:40:18.8 INT-2	okay. yeah.		
20:40:21.1 INT-1	so we'll just maintain here we're already at seven.		
20:40:23.6 INT-2	okay.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
		20:40:25.1 TWR-A	Muscle twelve contact Andrews Tower one one eight point four.
		20:40:28.1 MUSL12	Muscle twelve switching thank you.
		20:40:29.9 TWR-A	you're welcome. Bluestreak fifty one seventy two contact departure.
		20:40:33.4 TWR-A	correction Bluestreak fifty one twenty five contact departure.
		20:40:38.0 TWR-A	*.
		20:40:40.8 TWR-A	Bluestreak fifty one sixty four Washington Tower * circle runway three three * traffic on a eight mile final for runway one line up and wait. [transmission broken]
		20:40:50.9 TWR-A	* twenty three fifty five Washington Tower winds are three two direct one four gust two six traffic * * arrivals runway one. [transmission broken]
		20:41:07.0 TWR-A	Bluestreak fifty two forty seven when able left turn november * three three.
		20:41:14.5 TWR-A	Bluestreak * * thank you left turn november ground.

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
		20:41:19.8 TWR-A	Bluestreak fifty one sixty four winds are three two direct one five gust * traffic * for runway three three runway one clear for takeoff.
20:41:35.1 INT-1	alright Blackjack's already cleared the P's so he's not going to be a factor for us.		
20:41:40.0 INT-2	yeah.		
20:41:42.8 INT-1	and what did we request one to four? so we'll head straight down the river.		
20:41:47.1 INT-2	one for Hains...and...four		
		20:42:03.1 BLJK1	tower Blackjack one Memorial Bridge.
20:42:06.4 INT-1	*.		
		20:42:08.7 TWR-A	Blackjack one roger.
20:42:12.5 INT-1	I did not hear his route...after...		
20:42:14.5 INT-2	the Memorial.		
20:42:15.4 INT-1	yeah.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:42:16.1 INT-2	I don't see him yet.		
20:42:18.0 INT-1	* he's gonna be down in the lights.		
		20:42:19.4 TWR-A	<i>Bluestreak fifty one sixty four contact departure.</i>
		20:42:35.0 TWR-A	<i>** hold **.</i>
20:42:36.5 INT-1	* the Chain Bridge at seven hundred feet. good. next'll be Key Bridge we'll be down to three hundred.		
		20:42:42.3 TWR-A	<i>Bluestreak fifty three zero seven Washington Tower winds are three two direct one seven gust two five can you take runway three three?</i>
20:42:43.2 INT-2	I see it. to the left.		
		20:42:49.3 TWR-A	Bluestreak fifty three zero seven roger runway one cleared to land traffic * prior to arrival.
		20:43:09.5 TWR-A	Bluestreak fifty three forty two Washington Tower winds are three two direct one seven gust five can you take runway three three?
		20:43:20.0 TWR-A	Bluestreak fifty one seventy eight left turn at sierra ground point seven.

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
		20:43:39.6 TWR-A	<i>Bluestreak fifty three forty two at the Wilson Bridge change to * change to runway three three runway three three. [transmission broken]</i>
20:43:40.0 INT-2	gettin' choppy close to the ground.		
20:43:43.5 INT-1	oh yeah down low it's definitely gonna get choppy.		
20:43:45.8 INT-2	yeah.		
20:43:47.7 INT-2	we're at three hundred.		
20:43:49.8 INT-1	roger got you at four looking for *.		
		20:43:51.5 TWR-A	Blackjack one landing at boardwalk will be at your own risk when able report landing assured winds are currently three three direct one five gust two five.
		20:43:59.7 BLJK1	landing own risk and wilco for Blackjack one copy the winds thanks.
		20:44:15.5 TWR-A	* * traffic on * four mile final runway one line up and wait * ready to go.
20:44:27.7 INT-1	alright there's three hundred for two hundred.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
		20:44:29.2 TWR-A	<i>Bluestreak fifty three zero five currently number two for departure following an American seven three off your left.</i>
		20:44:35.7 TWR-A	Delta eight thirty two currently number three following an American C-R-J.
		20:44:41.8 TWR-A	* twenty three fifty five left turn november ground point seven good day.
		20:44:48.4 TWR-A	looks like you have november. let me know if you have it.
20:44:55.1 INT-2	two hundred.		
		20:44:56.9 TWR-A	alright. ground point seven good day.
		20:45:00.2 TWR-A	* seven * winds are three two direct one five gust two five * * no delays runway one.
		20:45:09.3 BLJK1	Blackjack one landing assured boardwalk good night.
		20:45:11.7 TWR-A	Blackjack one roger freq change approved good night.
		20:45:13.8 RDO-1	PAT two five Memorial.
20:45:14.5 INT-2	*.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
		20:45:16.1 TWR-A	PAT two five roger.
20:45:17.1 INT-2	alright.		
20:45:17.8 INT-1	lots of right pedal ma'am.		
20:45:19.0 INT-2	oh-kay.		
20:45:20.0 INT-1	there we go now we can make the turn.		
20:45:21.3 INT-2	okay.		
		20:45:22.7 TWR-A	<i>American thirty one (thirty/thirty one) Washington Tower winds are three one direct one four * runway one clear to land * *.</i>
20:45:22.7 INT-2	* * *.		
20:45:27.6 INT-3	clear left.		
20:45:28.8 INT-2	clear left. crane. no factor.		
20:45:32.4 INT-1	you're at three hundred feet. come down for me.		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:45:33.9 INT-2	yeah.		
20:45:36.0 INT-2	go down two hundred.		
20:45:50.6 INT-1	crane out the right. no factor.		
20:45:52.2 INT-2	okay.		
		20:45:55.0 TWR-A	* * contact * good day.
		20:45:57.9 TWR-A	American seventy one thirty * final.
		20:46:01.6 TWR-A	PAT two five traffic just south of Wilson Bridge is a C-R-J at one thousand two hundred feet for runway three three.
		20:46:07.9 RDO-1	PAT two five has the traffic in sight request visual separation.
		20:46:10.5 TWR-A	separation approved.
		20:46:11.6 TWR-A	American sixteen thirty tower runway one line up and wait traffic * runway three three this traffic on a six mile final.
		20:46:19.1 MUSL7	tower Muscle seven approaching Springfield request route three to zone six.

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
		20:46:24.5 TWR-A	seven stand by.
20:46:26.4 INT-1	ha ha.		
20:46:27.2 INT-2	ha.		
		20:46:29.8 TWR-A	Bluestreak fifty three zero seven keep rolling out to november ground point seven good day.
20:46:34.0 INT-1	he's got'em stacked up tonight.		
20:46:36.0 INT-2	(yeah/kinda) busy.		
20:46:40.4 INT-1	alright three three zero so now we've pretty much got a right quartering tailwind going to be pushing you.		
20:46:43.7 INT-2	yeah.		
		20:46:45.7 ARCR1	National Tower good evening AirCare one's with you a single medivac heli- [transmission cut off by next tower transmission]
		20:46:48.0 TWR-A	<i>American sixteen * one * one four gust two five traffic * * runway three three no delay runway one clear for immediate takeoff.</i>

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:46:48.9 INT-2	* crabbing.		
20:46:50.6 INT-1	yeah.		
20:46:51.8 INT-2	better not to fight the wind.		
20:46:56.6 INT-2	right pedal.		
		20:46:58.7 TWR-A	Muscle seven say request.
		20:47:00.8 MUSL7	and Muscle seven request route three to zone six.
		20:47:04.7 TWR-A	Muscle seven you said route three to zone six?
		20:47:06.7 MUSL7	a-firm Muscle seven.
		20:47:07.9 TWR-A	Muscle seven approved.
		20:47:09.1 MUSL7	Muscle seven *.
		20:47:15.0 ARCR1	National Tower good evening AirCare one's with you single medivac helicopter one thousand feet two niner niner one on the altimeter requesting zone six for Fairfax.

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
		20:47:25.5 TWR-A	medivac AirCare one Washington Tower nearest altimeter two niner niner zero approved through Washington class bravo airspace before landing assured at Fairfax.
		20:47:34.0 MEDI1	two niner two niner niner zero for AirCare one copies.
		20:47:39.1 TWR-A	PAT two five you have the C-R-J in sight? [sound of rapid beeping consistent with conflict alert audible in background]
		20:47:41.9 TWR-A	PAT [transmission interrupted by 0.8 second mic key from PAT-25] C-R-J.
		20:47:44.1 RDO-1	PAT two five has uh- aircraft in sight request visual separation.
		20:47:47.3 TWR-A	*.
20:47:47.8 INT-2	(woah/below).		
20:47:52.5 INT-1	alright kinda come left for me ma'am I think that's why he's asking.		
20:47:54.3 INT-2	sure.		
20:47:55.3 INT-1	we're kinda...		

Time and Source	Intra-Aircraft Communication	Time and Source	Over-the-Air Communication
20:47:55.4 INT-2	oh-kay. fine.		
20:47:56.4 INT-1	...out towards the middle.		
		20:47:57.6 TWR-A	<i>American four seventy two Washington Tower [transmission cut off by impact]</i>
20:47:57.7 INT-2	[sound similar to mumbling]		
		20:47:59.2 RDO	[sound similar to ELT signal]
20:48:00.1 FPS	[sound of rapid beeping similar to stabilator auto mode fail master caution, continues to end of recording]		
2048:04.4 END OF TRANSCRIPT END OF RECORDING			